

Week 1: Set Up Your AI-Course Portfolio Website

🌐 the-graph-courses.github.io/aiw_materials/week_01_workshop/week_01_workshop_instructions.html

This guide walks you through the steps to set up your AI-Course Portfolio Website for the Generative AI for Work & Research course.

⚠️ **NOTE:** This document is a short summary of the steps. For clearer instructions, consult the corresponding video on the website.

1 Open ChatGPT & switch to o3

- Go to chatgpt.com.
- Click the ▼ **Model** dropdown at the top left of the chat window.
- Select **OpenAI o3**.

Run out of o3 messages? That's fine. Most other models can handle a single page website.

2 Paste the prompt & course text

- Copy any relevant parts of the course description from the-graph-courses.org/aiw.
- Start a new message and create an instruction similar to this:

```
Build a one-page portfolio website for my "Generative AI for Work & Research" course.
```

```
Use the course information below and my CV.
```

```
At least these three sections:
```

- About me
- About the course
- My weekly submissions (here we will add links to each submission later on)

```
Show the HTML in a Canvas.
```

```
Avoid frameworks.
```

```
Don't include my email in the site to avoid spam
```

```
[PASTE THE COURSE TEXT AND YOUR CV HERE]
```

- (If you are very private, you can use a pseudonym or use a random sample CV from the internet)

3 Render the Canvas & refine

- ChatGPT should reply in a code block canvas.
- Click the **Run code** button at its top-right to open the live view.
- Now you can ask follow-up prompts. For example:
 - "Change the accent colour to green."
 - "Use a serif font for headings."
 - Upload a screenshot of a site you like → "Match this layout."

4 Stop preview & copy `index.html`

- Click the ■ **Stop** button (same location as the eye) to close preview.
- Select all the HTML inside the code box then copy it with **Ctrl/Cmd+C**.

5 Create a GitHub account (one-off)

GitHub is a platform for hosting and collaborating on code. It offers free hosting for static websites. We will use it to host your portfolio. - Visit github.com → click **Sign up** and follow the steps.

6 Make a new repository

- Click the + icon (top-right) → **New repository**.
- Name it `graph-ai-portfolio` (something similar).
- Leave it **Public**. Click **Create repository**.

7 Add your HTML file

- Inside the GitHub repo you created, find the button to create a new file and click it.
- Type `index.html` in the filename box, lowercase with no spaces.
- Paste the HTML you copied into the main code box below.
- Click **Commit new file**.

8 Turn on GitHub Pages

- Go to **Settings** (tab near top).
- In the sidebar, click **Pages**.
- Under *Source*, choose **Deploy from a branch**.
- Select **main** branch
- Click **Save**. Wait ~1 minute.
- Your site URL appears at the top (e.g. `https://username.github.io/ai-portfolio/`). Click to check.

9 Add a profile picture to GitHub

- In the repo, locate the button or icon to upload a file.
- Upload your image.
- Click **Commit changes**.
- Open the uploaded image → right-click it → **Copy image address**.
- Back in ChatGPT, ask the model to insert the image, which you paste as a link, as a circular avatar at the top.
- Copy the new HTML → return to `index.html` → click **Edit** (🖋️) → replace code → **Commit**.
- Refresh your live site – after about 1 minute the avatar should appear.

10 Submitting your work

- On the GitHub repo, click on the **index.html** file.
- Locate the download button or icon in the top right corner.
- Click it to download the file.
- Upload the file to the course portal.

11 Optional Challenge: Create a Professional CV with LaTeX

LaTeX is a powerful typesetting system commonly used for academic and professional documents. It's another markup format like HTML or Markdown, but specifically designed for creating beautifully formatted PDF documents. In this optional (ungraded) challenge, you'll create a professional CV using LaTeX.

11.1 Step 1: Get the LaTeX Template

- Go to overleaf.com → **New Project** → **Blank Project**.
- Click **Templates** (left) → search "simple resume" → **Use template**.
- Copy the entire LaTeX template code.

11.2 Step 2: Format Your CV with AI

- Take the LaTeX template to ChatGPT (using o3 model).
- Paste your CV text and the LaTeX template.
- Ask the model to reformat your CV using the LaTeX template structure.
- Copy the AI-generated LaTeX code.

11.3 Step 3: Create Your CV

- Go back to Overleaf and paste the AI-generated LaTeX code.
- Click **Recompile** to see your formatted CV.
- Make any necessary adjustments.
- Click **Download PDF** to save your CV.

11.4 Step 4: Add to Your Portfolio

- Upload your CV PDF to Google Drive.
- Get a shareable link (make sure it's publicly accessible).
- Ask ChatGPT to add a "Download CV (PDF)" button to your website, linking to your Google Drive URL.
- Update your [index.html](#) with the new code.
- Commit the changes to GitHub.

Workshop 2: Automated Data Analysis & Reporting

 the-graph-courses.github.io/aiw_materials/week_02_workshop/week_02_workshop_instructions.html

The Graph Courses

Welcome to another session of your Generative AI Workshop! In this session, you'll use Data Analysis AI tools to compile a brief report (HTML or PDF) analyzing the relationship between two chosen country indicators from the Gapminder data repository here:

<https://gapminder.org/data>.

Your final report should contain:

1. A world map visualization of your first selected indicator (for the chosen recent year).
2. A world map visualization of your second selected indicator (for the chosen recent year).
3. A scatterplot showing the relationship between your two chosen indicators.
4. A description and interpretation of the relationship between the two indicators. This may include a correlation coefficient (e.g., Pearson's r) if appropriate, or a qualitative description based on the scatterplot.
5. A short reflection written by you about the process of interacting with the AI tools.
E.g. What was easy? What was difficult? What was surprising? What did you learn?

Note that rather than grading the technical precision of the report (we understand that you are not all data analysts), we will simply check that you have followed the overall instructions and written an honest reflection.

Tool Tracks

You will choose one of the following tools based on your coding experience:

- **Students WITH significant coding experience in R:** You should use **Cursor**, an AI-powered code editor, with R.
- **Students WITHOUT significant coding experience:** You will use **Julius**, a more user-friendly AI data analysis tool with a chat interface.
- **If you finish your assigned tool track early, you are encouraged to try the other one!**

Part 1: General Setup & Data Selection (All Students)

1. Choose Your Data:

- Go to Gapminder's data repository: <https://gapminder.org/data>
- Browse the indicators and **select TWO indicators** that you hypothesize might be related.

Think about what story you want to explore. For example, a famous relationship is GDP per capita vs. Life Expectancy. Consider other potential meaningful connections.

- For each chosen indicator, download the data as a CSV file.

2. Data Preparation Note: Data will be subsetted to a single year:

After loading your data into Cursor or Julius, you will ask the AI to **subset it to the most recent year that has good data coverage**. "Good data coverage" means not too many missing values. When browsing on Gapminder, try to identify a specific recent year you'd like to focus on that has good coverage for both your chosen indicators, and make a note of this year.

Part 2: Tool-Specific Instructions

Please follow the instructions for the tool assigned to your group.

Section A: For Students WITH Coding Experience (Using Cursor with R)

You will use Cursor to generate an R Markdown report.

1. Setup Cursor:

- You'll need to set up Cursor and can use their free 14-day trial.
- **Watch our setup video before you begin:** [AI Automated Data Analysis with R and Cursor \(for R users only\)](#)

2. Project folder:

Create a new project folder on your computer. Place the two CSV files you downloaded from Gapminder into this folder.

3. Generating Your Report in Cursor:

- Open your project folder in Cursor. Create a new R Markdown (`.Rmd`) file with an appropriate name.
- Before using the Agent, it's recommended to change the model to **Claude 3.7 Sonnet** for better performance with R. You can usually find this setting in a dropdown menu associated with the chat or agent interface.
- Then use Cursor's "Agent Mode" to help you write the R code and narrative.
- **Initial Agent Prompt Idea:** Here is an example of an initial agent prompt:

```
You are an AI agent that creates R data reports.  
You create reports by writing code to an Rmarkdown file (with  
github_document output), knitting that file, then reading the output md  
file to validate your work.  
You repeat this edit, knit, review loop until the reporting task is  
achieved.  
Use plots, tables, and inline R code where appropriate.  
Code should be hidden from the output.  
Your task is further described below.
```

- **Specific Task Prompting Example:**

```
Now create a data report in my Rmd file using the two datasets I've  
provided ([indicator1] and [indicator2]).
```

```
The report should:
```

1. Load both datasets.
 2. For each indicator, subset the data to ``[CHOSEN_YEAR]``.
 3. Create a world map visualization for the first indicator.
 4. Create a world map visualization for the second indicator.
 5. Merge the data for the two indicators.
 6. Create a scatterplot showing the relationship between these two indicators.
 7. Provide a brief interpretation of the relationship shown in the scatterplot (you can calculate Pearson's correlation coefficient `$$r$$` if it's helpful).
- ```
Ensure all plots are clearly titled and labeled.
```

- **Iterate and Refine:** Work with the agent to refine the figures and text until you are happy with the report. You can ask it to change the title of a plot, or to use a different color scheme, for example.
- **Hallucinations related to plots:** Cursor cannot see your plots. It may hallucinate interpretations of the plots that are inaccurate. You can ignore these.

- **A Note on AI Interaction:**

- If the AI seems to be going off track or taking too long, look for a “stop” button (often near the prompt input box) to interrupt it.
- Often, if you don’t like the output, it’s more effective to go back to your previous prompt, edit it to be clearer or more specific, and resubmit, rather than trying to correct the AI with a long series of follow-up messages. Longer conversations can sometimes confuse the AI.

- **Troubleshooting R Package Installations:**

If Cursor has persistent trouble installing R packages, you might need to install them manually. Create a temporary R script (`.R`) file in your project. In this script, use the `install.packages("package_name")` command for each required package. Run this script. R might ask you to choose a CRAN mirror (a server to download from); pick any. Once the packages are installed, you can delete the temporary R script.

4. **Add Your Reflection:** Before your final knit, make sure to add your short reflection (point 5 from “Your final report should contain”) directly into your R Markdown file. This should be part of the narrative of your report.
5. **Final Output:** Your primary submission will be the HTML file generated from knitting your R Markdown file. (That is, while your AI agent will output the report as an `.md` file while it is working, you should change the output format when you are done and ready to submit. To do this, in your `.Rmd` file, change `output: github_document` to `output: html_document` in the header section before your final knit.)

## Section B: For Students WITHOUT Prior Coding Experience (Using Julius)

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You will use Julius, an AI data analysis chat-based tool, to analyze your data and generate content for your report.

1. **Setup Julius:**

- Go to [Julius.ai](https://julius.ai) and sign up.
- For a 2-week trial of premium features, go to your **Settings**, then **Usage**, and under **Institutional Codes**, enter: `GRAPHAIR`
- **Please do not share this code.**

## 2. Starting Your Analysis in Julius:

- Start a new “thread” in Julius.
- **Select your tools:**
  - **Code Language:** Julius can use Python or R. If you have no preference, we recommend selecting **R**. The default figures from R are often a bit nicer, and more members of The Graph team are R experts, so we can better assist with debugging if needed. The code language changer is usually found at the top left of the Julius window.
  - **AI Model:** It’s recommended to switch to the **Claude 3.7 Sonnet** model for potentially better results. The model selection is typically in a dropdown menu below the prompt input box.
- Now, **upload your two CSV files** to the conversation.

## 3. Chatting with Julius (Example Prompt):

- **Chat with Julius Example Prompt:**

I am writing a report on the relationship between the indicators [indicator1] and [indicator2]. Both datasets have been added.

Please help me do the following:

1. Load both datasets.
  2. For each indicator, subset the data to `[CHOSEN\_YEAR]`.
  3. Create a world map visualization for the first indicator.
  4. Create a world map visualization for the second indicator.
  5. Now merge the data for the chosen indicators.
  6. Create a scatterplot showing the relationship between these two indicators.
  7. Provide a brief interpretation of the relationship shown in the scatterplot (you can calculate Pearson's correlation coefficient  $r$  if it's helpful).
- Make sure the plots have good titles and labels.

- **Review and Refine:** Julius will show its steps, data, and plots. Work with it to refine the figures and text until you are happy with the output plots. You can ask it to change the title of a plot, or to use a different color scheme, for example.
- **A Note on AI Interaction:**
  - If the AI seems to be going off track or taking too long, look for a “stop” button (often near the prompt input box) to interrupt it.
  - Often, if you don’t like the output, it’s more effective to go back to your previous prompt, edit it to be clearer or more specific, and resubmit, rather than trying to correct the AI with a long series of follow-up messages. Longer conversations can sometimes confuse the AI.



#### 4. Creating Your Report Document:

- **Download Outputs:** Download the individual plots (as images, e.g., PNG) that Julius generates. You will also copy any key text outputs, such as data summaries or interpretations.
- **Create a DOCX/GDocs file:** Open a word processor (like Microsoft Word, Google Docs, or Pages) and create a new document.
- **Assemble initial report content:**
  - Add a title, your name, and the date.
  - Insert the plots you downloaded.
  - Paste in the relevant text outputs from Julius.
  - Organize this content into a clean and readable report.

#### 5. Add Your Reflection:

Before finalizing your report, add your short reflection (point 5 from “Your final report should contain”) to your document.

#### 6. Final Output:

**Export as PDF** from your word processor. This PDF is your primary submission.

### Part 3: Submission

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Upload your final report file (HTML or PDF) to the designated submission link on the course workshop page.

### Part 4: Optional - Add to Your GitHub Pages Portfolio

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As a final step, you can add your report to your portfolio website which you created earlier in the course.

1. **Upload your Report:** Add your final report file (HTML or PDF) to the GitHub repository that hosts your portfolio website.
2. **Get the Link:** Once uploaded, navigate to the file in your GitHub repository and copy its direct link.

### 3. Update Your Portfolio with ChatGPT:

- Go back to the ChatGPT conversation you used to build your portfolio website. (If you don't have it, you can start a new one, provide ChatGPT with your existing `index.html` file content, and then give it the new instructions).
- Ask ChatGPT to add a new entry for this workshop. For example:

Please update my portfolio website. I want to add a new project for "Workshop 2: Automated Data Analysis".  
Include a brief description like "Analyzed Gapminder data to explore relationships between two global indicators using AI tools."  
Then, add a link to my report: [paste the direct link to your uploaded report file here]

- ChatGPT will provide you with updated `index.html` code. Replace the content of your `index.html` file in your GitHub repository with this new code. Commit and push the changes.
- Verify that your live GitHub Pages site now shows the new entry and the link works.

# Analyzing Job Market Data with LLMs in Sheets and AI Reporting Agents

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 [the-graph-courses.github.io/aiw\\_materials/week\\_03\\_workshop/week\\_03\\_workshop\\_instructions.html](https://the-graph-courses.github.io/aiw_materials/week_03_workshop/week_03_workshop_instructions.html)

The Graph Courses

Welcome! In this assignment, you'll use Large Language Models (LLMs) directly within Google Sheets to extract specific information from hundreds of data analysis job postings. (This is the kind of analysis that we do at Graph Courses to inform our course offerings.)

Afterwards, you'll take this extracted data and use an AI-powered analysis tool (Julius or Cursor, based on your experience from the previous workshop) to create visualizations and a concise report.

## Prerequisites

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**Before you begin:** This assignment assumes you have completed the pre-work, which covers setting up and using LLMs in Google Sheets with OpenRouter. If you haven't, please briefly review the video here: [Using LLM APIs with OpenRouter and Google Sheets](#) to ensure you have the necessary context. You don't have to code/type along with the video; a brief review is enough.

Then come back to this assignment.

## Your Final Report Structure

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Your submitted report for this assignment should include the following sections:

1. A brief introduction (1-2 sentences) describing the data and the project's aim.
2. Analysis of "Years of Experience" (including a visualization and your interpretation).
3. Analysis of "Programming Language Requirements" (including a visualization and your interpretation).
4. A personal reflection on the process (see details in Part 2).

(In your interpretations, be sure to note the limitations of the small sample size.)

## Part 1: Data Extraction in Google Sheets

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You will begin by setting up Google Sheets to call an LLM API to extract specific information from job descriptions.

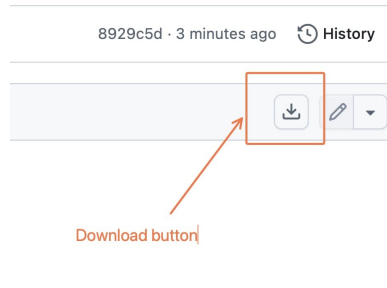
### Part 1A. Setting Up Your Google Sheet

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## 1. Get the Data:

The dataset for this assignment is a CSV file of Data Analyst job postings: [data\\_analyst\\_postings\\_sample.csv](#). There is a download icon at the top right of the file

Screenshot:



(This data was originally scraped from Glassdoor in June 2020. The scraping project and full dataset can be found here: [Original Data Source](#))

## 2. Import Data into Google Sheets:

- Create a new Google Sheet and give it a name like “Data Analyst Job Market Analysis”. (You can quickly create a new sheet by going to “sheets.new” in your browser.)
- Go to **File** → **Import**.
- In the Import file dialog, select the **Upload** tab.
- Drag the downloaded [data\\_analyst\\_postings\\_sample.csv](#) file into the upload area. The default import settings should be fine.
- **Adjust Row Height (if needed):** If the rows are too tall due to long job descriptions, select all cells (press Ctrl+A or Cmd+A two times), then move your mouse to the divider line between any two row numbers on the left (e.g., between row 1 and row 2). Your cursor will change. Click and drag the divider upwards to resize all selected rows.

## 3. Set Up Your API Key:

In your Google Sheet, choose an empty cell that’s out of the way (e.g., **M2**). Paste your personal OpenRouter API key into this cell.

#### 4. Add the **MYGPT** Apps Script Function:

- Go to **Extensions** → **Apps Script**.
- Replace any starter code in the editor with the exact snippet below:
- Click the “Save project” icon ().
- Click “Run” once. Google will ask you to authorize the script. Follow the prompts to allow it. You might see a “This app isn’t verified” warning; you can proceed by clicking “Advanced” and then “Go to (your project name)”.
- You may need to go through the authorization process more than once. You will know it has worked when you see “Execution completed” after pressing “Run”.

## Part 1B. Extracting Years of Experience Required

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You’ll use the LLM API to find the minimum years of experience mentioned in the **Job Description** column.

1. **Create New Column:** In your sheet, find the first empty column to the right of your data. Add a header like **Minimum Years Experience**.
2. **Formulate the Prompt & Apply MYGPT:**
  - We will use the **google/gemini-2.5-flash-preview** model.
  - Assuming your first job description is in cell **C2** (adjust if it’s in a different column for your imported data) and your API key is in cell **M2**, the formula in the first cell under your **Minimum Years Experience** header would be:

```
=MYGPT(CONCATENATE("Extract the minimum years of experience required from
this job description. Return ONLY a number. If a range is given, return the
lower number. If no years of experience are explicitly mentioned, return
'NA'. Here's the job description: ", C2), "google/gemini-2.5-flash-preview",
M2)
```

Test this on a few rows (e.g. 2-5) and manually verify that the outputs are reasonable.

3. **Apply to All Rows:**

Once you have tested the formula on a few rows, drag the fill handle (the small square at the bottom-right of the selected cell) down to apply it to all job descriptions. This will make an API call for each row, so it might take a moment.

4. **Lock in Results:** Once the LLM has processed the column and you have the results:

- Select the column of results and copy them (Ctrl+C or Cmd+C).
- Right-click on the column header again, then **Paste special** → **Values only**. This converts the formulas to static text, preventing accidental re-runs and saving API costs.

## Part 1C. Extracting Programming Language Requirements

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Next, you'll ask the LLM to identify if job descriptions mention R, Python, both, or neither.

1. **Create New Column:** Add another new column with a header like **Data Languages**.
2. **Craft Your Own Prompt:**
  - For this task, you need to write the prompt yourself.
  - Your prompt must clearly instruct the LLM to analyze the **Job Description** and return **ONLY one of these four specific strings** based on the data programming language requirements in the job description: R, Python, Both, or Neither.\*\*
  - **Hint:** Be very explicit. A good prompt will state the options clearly and tell the model to choose only one.
3. **Apply MYGPT Function:**
  - Use your crafted prompt with the **MYGPT** function, similar to Task 1, concatenating it with the relevant job description cell (e.g., **C2**) and using **google/gemini-2.5-flash-preview** and your API key reference (e.g., **\$M\$2**).
  - Apply to a few rows first to validate that the model is doing what you want, then to all rows.
4. **Lock in Results:** Again, once processed, select all cells with results in your **Data Languages** column, copy them, and then paste special as values only.

## Side Note: Is An LLM Needed just to extract 'R' and 'Python'?

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A valid question at this point is "Is an LLM needed just to extract R and Python? Can't I just use a simple text search?" While this would work for Python, it would not work for R, unfortunately.

The challenge is that "R" is an extremely common letter in English text. A simple text search for "R" in job descriptions would return a lot of false positives like "R & D (research and development)", "H R (human resources)", "RA (research assistant)" and others.

An LLM is more robust because it understands context, allowing it to accurately distinguish between what capital R means in "R programming" and what it means in "HR department".

## Part 2: Reporting with AI Agents (Julius or Cursor)

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Now that you've extracted data in Sheets, you'll use an AI agent for visualization and reporting.

Ensure your Google Sheet has the original **Job Description** and **Median Salary Estimate USD** columns, plus your newly created and value-pasted **Minimum Years Experience** and **Data Languages** columns.

Next, in Google Sheets, go to **File → Download → Comma Separated Values (.csv)**. Save this file to your computer.

You will use either Julius or Cursor, the tool you used in a previous workshop, to analyze your downloaded CSV and generate your report.

**1. Tool Choice:**

- **Julius:** If you prefer a chat-based interface or have less coding experience. Remember to switch to the Claude 4 sonnet model.
- **Cursor:** If you have coding experience (e.g., R or Python) and prefer an AI-assisted code editor.
- Refer to the instructions from our previous workshop if you need a refresher on setting up or basic interaction with your chosen tool.

## 2. Generate Your Report:

- Upload your downloaded CSV file to Julius (Or for Cursor, create a new folder, open that in Cursor, then add the CSV file to that folder).
- Your goal is to create a report document that includes:
  - A brief introduction (1-2 sentences) describing the data and project.
  - **Years of Experience Analysis:**
    - A scatter plot: **Minimum Years Experience** (x-axis) vs. **Median Salary Estimate USD** (y-axis).
    - A brief written interpretation of the plot.
  - **Programming Language Analysis:**
    - A box plot: **Median Salary Estimate USD** distributions for each category in **Data Languages**.
    - A brief written interpretation of the plot.
  - **Reflection:** A short reflection (e.g., 1-2 paragraphs) written by you (not the AI) on your experience with this assignment. Consider:
    - What was easy or difficult?
    - What was surprising or noteworthy?
    - What did you learn?
- **Example Prompting Idea for JULIUS (adapt as needed):**

I've added a CSV which contains data from job postings. It contains job data with columns like 'Median Salary Estimate USD', 'Minimum Years Experience ' (numeric, may contain 'NA' or non-numeric text if conversion failed), and 'Data Languages' (categorical: "R", "Python", "both", "neither").

Help me generate the following figures:

- A scatter plot of 'Minimum Years Experience ' vs. 'Median Salary Estimate USD'.
- A box plot of 'Median Salary Estimate USD' grouped by 'Data Languages'.

I will compile these into a report and add my own introduction and reflection.



- **Example Prompting Idea for CURSOR (adapt as needed):**

You are an AI agent that creates data reports.

- You create reports by writing code to an Rmarkdown file (with github\_document output), knitting that file, then reading the output md file to validate your work.
- You repeat this edit, knit, review loop until the reporting task is achieved.
- Use plots, tables, and inline R code where appropriate.
- Code should be hidden from the output.

Now, I've added a CSV which contains data from job postings. It contains job data with columns like 'Median Salary Estimate USD', 'Minimum Years Experience ' (numeric, may contain 'NA' or non-numeric text if conversion failed), and 'Data Languages' (categorical: "R", "Python", "both", "neither").

Help me generate a report with the following figures:

- A scatter plot of 'Minimum Years Experience ' vs. 'Median Salary Estimate USD'.
- A box plot of 'Median Salary Estimate USD' grouped by 'Data Languages'.

I will compile these into a report and add my own introduction, interpretation, and reflection.

## Side Note: Why This Multi-Step Process?

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You might wonder: “Why not just upload the original, raw CSV of job postings directly into Julius, Cursor, or even ChatGPT and ask it to do everything?”

While powerful, LLMs and AI agents have limitations, especially with large, unstructured inputs:

- **Context Window Limits:** LLMs can only process a certain amount of text (the “context window”) at once. A large CSV with many long job descriptions can easily exceed this limit. The model might then ignore parts of your data or fail to process it comprehensively.
- **Focus and Accuracy:** Asking an LLM to perform many complex tasks (like reading hundreds of descriptions, identifying specific data points within each, cleaning them, and then visualizing) all from one massive, raw input can lead to confusion, errors, or “hallucinations.” The model might struggle to keep all instructions and data points straight.

## Part 3: Submission

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- **What to Submit:** Your final report as a **single PDF or HTML document**. This PDF or HTML should include all the sections outlined at the beginning: Introduction, Years of Experience Analysis (plot & interpretation), Programming Language Analysis (plot & interpretation), and your AI Reflection.

- **You don't need to submit your Google Sheet.** That is for your own work.

## Part 4: Optional - Add to Your GitHub Pages Portfolio

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You're encouraged to add this report to your GitHub Pages portfolio.

1. **Upload Your Report:** Add your final report file to the same GitHub repository that hosts your portfolio website.
2. **Get the Link:** Once the report is uploaded to your GitHub repository, navigate to the file and get its direct link. Usually, you can right-click the file and "Copy Link Address".
3. **Update Your Portfolio:**
  - You can ask ChatGPT (or a similar LLM) to help you update your `index.html` file. If you have your previous ChatGPT conversation for building the site, you can continue it. Otherwise, you might start a new conversation, provide the content of your existing `index.html` file, and then give your update request.
  - ChatGPT will provide updated `index.html` code. Carefully replace the entire content of your `index.html` file in your GitHub repository with this new code.
  - Commit and push the changes to GitHub.
  - Verify that your live GitHub Pages site shows the new entry and that the link to your report works correctly. This may take a few minutes to update.

# Portfolio Website Chatbot with Botpress

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 [the-graph-courses.github.io/aiw\\_materials/week\\_04\\_workshop/week\\_04\\_workshop\\_instructions.html](https://the-graph-courses.github.io/aiw_materials/week_04_workshop/week_04_workshop_instructions.html)

The Graph Courses

## Introduction

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Welcome! In this assignment, you'll create an LLM-powered chatbot using Botpress and embed it into your portfolio website that you created in a previous session.

Your completed chatbot will be able to:

1. **Answer CV/Resume Questions** - Provide information about your background using uploaded CV data
2. **Share Course Information** - Provide info about the Graph Courses GenAI syllabus and curriculum details
3. **Collect Visitor Information** - Gather names and emails from interested visitors for showcase day registration

### 1. Create Your Botpress Account:

- Go to [botpress.com](https://botpress.com)
- Sign up for a free account using your email (Note: Botpress provides \$5 in free credits, which should cover over 1000 messages on the default models. But for serious use, you will need to add credit to your account)
- Verify your email address if needed

### 2. Create a New Bot:

- Once logged in, click "Create Bot"
- Name your bot something like "Portfolio Assistant" or "[Your Name] Course Bot"
- Follow the instructions to open the bot in the studio

Your bot will use an "Autonomous Node", which means you can define your AI agent's purpose and behavior in just a few lines of text.

### 3. Set Bot Instructions:

In the main configuration area, find the “Instructions” field

#### Instructions

##### ## Identity

You are the Customer Support AI Agent for [COMPANY NAME]. You handle all customer inquiries, and provide assistance with common support tasks.

##### ## Scope

Enter instructions like:

You are a portfolio assistant for a Generative AI course student. Your main tasks are:

- Answer questions about the student's CV and background using the knowledge base
- Provide information about the Graph Courses GenAI syllabus and curriculum based on your knowledge base
- After answering a user's question, politely ask them whether they might be interested in our upcoming showcase event on Friday June 27th at 3pm GMT or Sunday June 29th at 5pm GMT
- If they express interest, you will collect their name and email for registration and record those in the Attendees table tool.

Be friendly, professional, and informative. Always try to be helpful while staying focused on these core topics. For other topics, politely decline to answer.

## Part 2: Set up the Knowledge Base

---

### 1. Navigate to Knowledge Base:

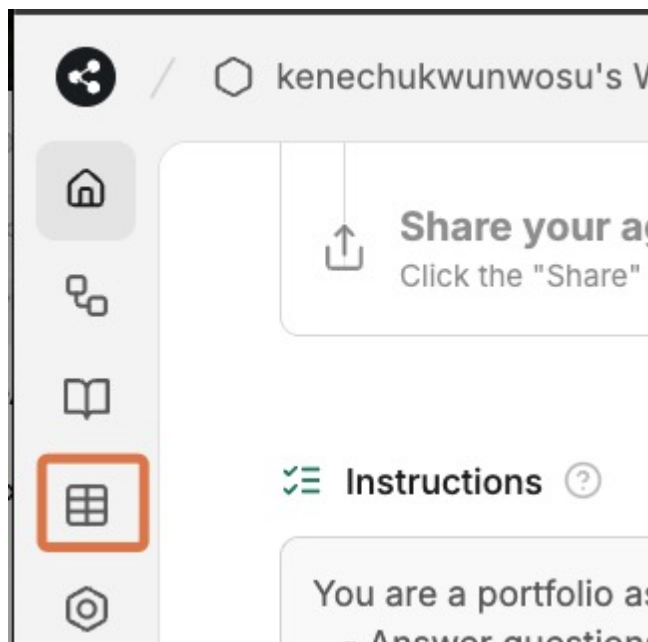
- Find the “Knowledge Base” section under the “Instructions” section
- Drag and drop your CV into the upload box. (If you don't want any of your personal information online, you can use a fake example CV from the internet, like here: <https://www.careers.ox.ac.uk/files/traditional-cv-examplepdf>)
- Download the syllabus from: [https://drive.google.com/file/d/1NV-Mo2UgXQwdTu\\_Jv-HtCVnB9Pf2Grng/view](https://drive.google.com/file/d/1NV-Mo2UgXQwdTu_Jv-HtCVnB9Pf2Grng/view), and upload it to the knowledge base.

## Part 3: Set Up Data Collection

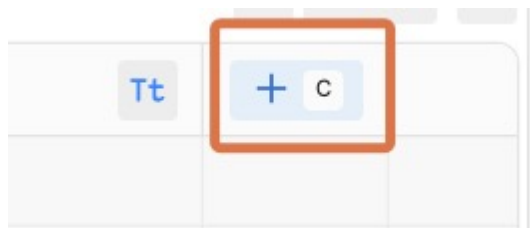
---

## 1. Create Attendees Table Structure:

On the Botpress sidebar, click on the Tables icon.



- Click on the New Table button at the top left and call it something like “Attendees”, then click to Save
- Click the new column icon to add the following two columns, leaving all settings as default
  - Name
  - Email



## 2. Configure Collection:

Return to the main configuration area, by clicking on the “Main” icon on the left sidebar



- Scroll down to the Tools section under Knowledge Base
- Click on the Add Tool plus icon
- Click on Tables then the Attendees table, then “Create Rows”. This gives your bot the ability to create rows in that table.

## Part 4: Testing and Publishing Your Bot

---

### 1. Use the Test Chat:

- The Botpress interface has a test chat panel on the right side of the screen
- Try asking questions about:
  - Your background (“Tell me about the educational background of [your name]”)
  - Course content (“What topics are covered in the GenAI course?”)
- The bot should interrupt at some point to ask if you might be interested in the event.
- Give your email and name, then check the Attendees table to see if the bot was successful in adding your information.

## 2. Publish:

- Click “Publish” button in the top-right corner
- Wait for the publishing process to complete

## 3. Get Embedding Code:

- After publishing, go to “Share”, also in the top-right corner
- Select “Configure”
- Find the “Chat Bubble” section
- Copy the two script lines that will look similar to this:

```
<script src="https://cdn.botpress.cloud/webchat/v2.4/inject.js"></script>
<script src="https://files.bpcontent.cloud/2025/05/27/14/20250527145803-
WFVSK7BL.js"></script>
```

# Part 5: Embedding Your Bot on Your Website

---

## Part 5A: Add Bot to Your Portfolio

---

### 1. Access Your GitHub Repository:

- Go to your GitHub Pages repository where your portfolio is hosted
- Open the `index.html` file for editing

### 2. Insert the Bot Code:

Paste the two script lines you copied from Botpress right before the closing `</body>` tag. This will be near the end of the document.

If you do not feel comfortable making this change yourself, you can ask an LLM to do it for you.

### 3. Commit and Deploy:

- Save your changes to the `index.html` file
- Commit the changes to your GitHub repository
- Your live site will update automatically (this may take 3-5 minutes)
- You can track the status of your deployment by clicking on the “Actions” tab in the menu of the GitHub repository page
- NOTE: Even after the site is done deploying, you may need to refresh the page a few times to see the bot appear, because your browser may be loading a cached version of the page.

## Part 5B: Test Live Integration

---

### 1. Visit Your Live Site:

- Go to your GitHub Pages URL
- Look for the chat bubble (usually appears in the bottom-right corner)

### 2. Test Full Functionality:

- Try the same tests you did before, but now on your live website
- Ensure the bot loads properly and responds correctly
- Verify the registration flow works in the live environment

### 3. Making changes to your bot:

WATCH OUT: If you make any changes to your bot, be sure to click on “Publish” again in the top-right corner to save your changes.

## Part 6: Optional - Customize the Appearance of Your Bot

---



1. **Navigate to the Customize Dashboard:** To customize the appearance of your bot, again click on the “Share” button in the top-right corner then select “Configure”. Make sure you are in the “Chat Bubble” section, then under the “Appearance and Theme” section, click on “Edit on Dashboard”.

**Share Bot** [Preview Bot](#)

Preview your bot live in the Botpress Webchat. You can also share it with friends by copying the shareable link below.

<https://cdn.botpress.cloud/webchat/v2.4/shareable...>

**More sharing options**

Embed the Botpress Webchat on your website or app in different ways.

[Configure](#)

**Chat Bubble**  
Web page corner

**Embed**  
Iframe on page

**Standalone**  
Mobile-friendly

**React**  
React library

**Script**

Add this script to your website to enable the chat bubble. Add the script in the `<head>` section of your website. [Documentation](#)

```
<script src="https://cdn.botpress.cloud/webchat/v2.4/inject.js"></script>
<script src="https://files.bpcontent.cloud/2025/06/01/17/20250601172542-F3HL7V62.js"></script>
```

**Webflow** **Wix** **Wordpress**

**Appearance and Theme**  
Customize the appearance of the chat bubble

[Edit on Dashboard](#)

2. **Customize the Appearance:** On this page, under the “Theme” section, you can change the color and font, and under the General section you can change the name of the bot, a description, an icon and a composer placeholder.

## Part 7: Submission

---

To submit:

- Upload a screenshot of your chatbot 'in action' on the graph courses website in the upload box.
- Paste a link to your website in the comments section.

# Workshop: Automation with Make.com

---

 [the-graph-courses.github.io/aiw\\_materials/week\\_05\\_workshop/week\\_05\\_workshop\\_instructions.html](https://the-graph-courses.github.io/aiw_materials/week_05_workshop/week_05_workshop_instructions.html)

The Graph Courses Team

In this workshop, you'll create an automated workflow using Make.com that processes consultation inquiries submitted through a Google Form. The automation will:

1. Monitor a Google Form for new submissions
2. Use an AI model to research the potential client's company and generate a report
3. Create a formatted Google Doc with the research findings
4. Send you an email notification with the report

The goal here is to build an extremely simple automation that demonstrates the power of connecting different web services with an AI Large Language Model (LLM). Setting up this automation is quite a bit of work, but the beauty of automations of course is that (in theory) you should only need to set it up once!

In the wild, you will find much more complex automations. Many organizations run a large fraction of their processes using automations on services like Make.com. With the rise of LLMs, these automations can do increasingly complex tasks.

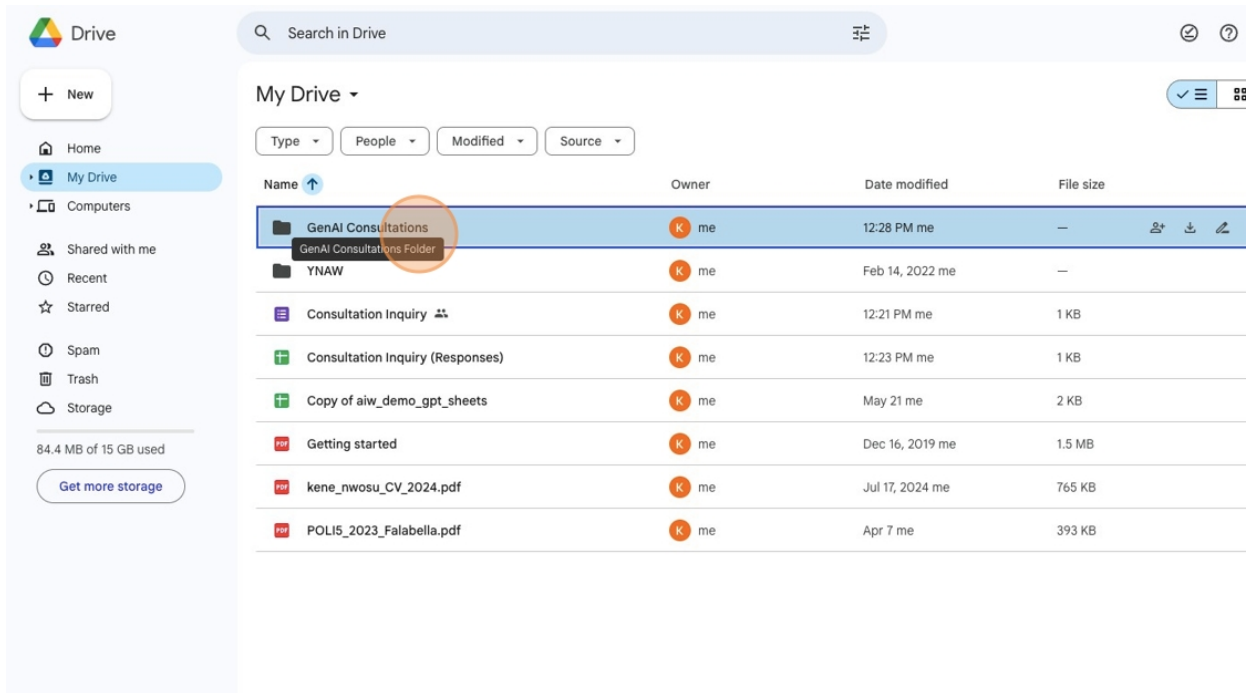
## Part 1: Setting Up Your Google Form and Data Collection

---

### Create Your Project Folder

---

1. In your browser, navigate to [drive.google.com](https://drive.google.com)
2. Click **New** at the top left → then click **New folder**
3. Name the folder something like **GenAI Consultations**
4. Double-click to open the folder



## Build Your Consultation Form

1. Inside the folder, again click **New** then click on **Google Forms** to create a new form.
2. In the title field enter: **Consultation Inquiry**
3. Set up the following form questions:
  - **Question 1:** "Name" (Short answer)
  - **Question 2:** "Email" (Short answer)
  - **Question 3:** "What is your company name and primary geographic area?" (Short answer)
  - **Question 4:** "What would you like help with?" (Paragraph)

**Consultation Inquiry** All changes saved in Drive

Questions Responses Settings

**Consultation Inquiry**

Let's discuss your needs. Fill out this quick form for a free consultation

**Name**

Short answer text

**Email**

Short answer text

**What is your company name and primary country? (To better understand your organization context)**

Short answer text

**What would you like help with?**

Paragraph

4. When done, click **Publish** at the top right to make your form live.

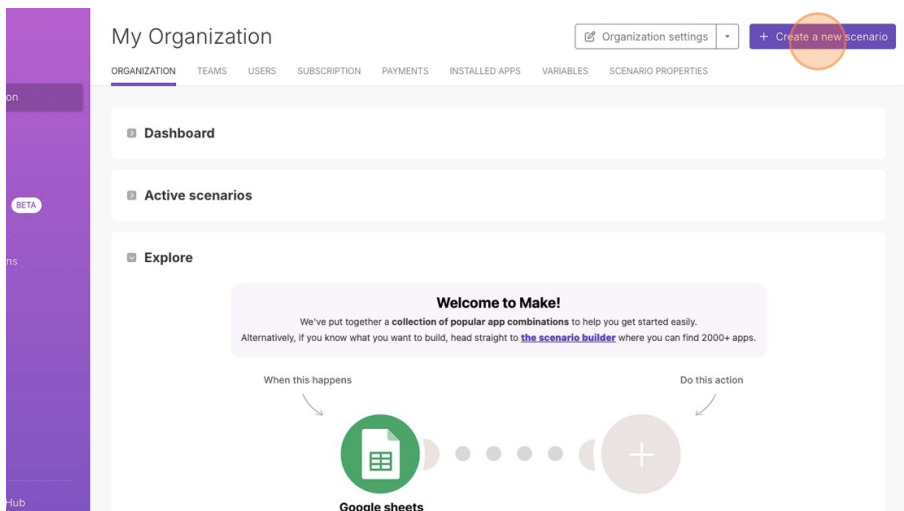
## Test Your Form and Connect to Sheets

1. Click the Share button at the top right, then copy the responder link.
2. Open the form in a new tab and submit a test response. For example
  - Name: John Doe
  - Email: john.doe@example.com
  - Company: The Graduate Institute, Geneva
  - Help needed: General advice on how to use GenAI to help students
3. Return to the form editor tab and click **Responses**. This will show you the responses to your form. You should see a new response in the list.
4. Click **Link to Sheets** → **Create** to generate a connected spreadsheet.
5. Share the spreadsheet by clicking **Share** in the top right → **Copy link**. Copy this link to your clipboard. We will use it in the next section.

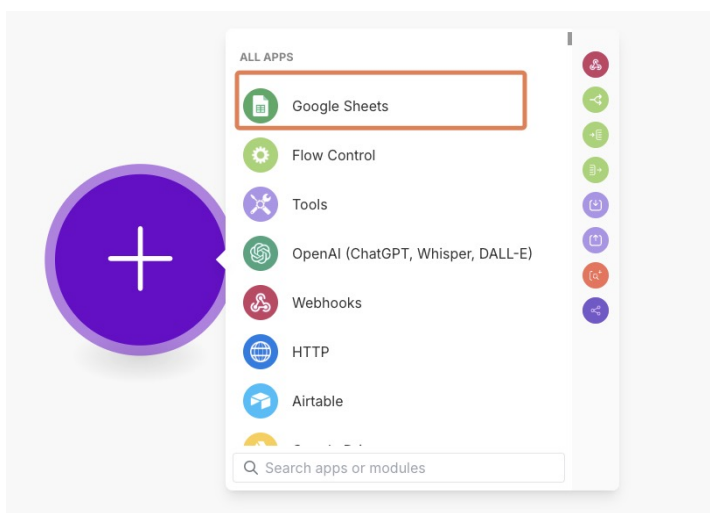
## Part 2: Setting Up Make.com Automation

### Build Your First Module: Google Sheets Watcher

1. If you have not yet done so, create a Make.com account. Once done, navigate to <https://www.make.com/en> and sign in.
2. Click **Create a new scenario**



3. Click the plus button to add a new module, and select **Google Sheets**



4. Choose **Watch New Rows ACID**. This will watch for new rows in the spreadsheet, and trigger when someone fills out the connected form.
5. Click **Creating a connection...** → **Sign in with Google**
6. You may need to select your Google account and authorize access. Give all requested permissions. (Note that you it is only *your* automations that have the access, not the entire Make.com service, so you do not need to worry about data security or leakage.)

7. Now, configure the module settings:

- **Spreadsheet ID:** Navigate to and select your consultation form spreadsheet. It should be called something like Consultation Inquiry.
- **Sheet name:** Select "Form Responses 1"
- **Limit:** Set to **1** (processes one response at a time)
- Click **Save**

Google Sheets

☒ **Connection \***

 My Google connection (ken...

 For more information on how to create a connection to Google Sheets, see the [online Help](#).

☒ **Search Method \***

Search by path

☒ **Drive \***

My Drive

☒ **Spreadsheet ID \***

/ Consultation Inquiry (Responses)

☒ **Sheet Name \***

Form Responses 1

☒ **Table contains headers \***

Yes

☒ **Row with headers \***

A1:Z1

 Enter the range of the table headers. E.g. A1:F1.

☒ **Limit \***

1

 The maximum number of results to be worked with during one execution cycle.

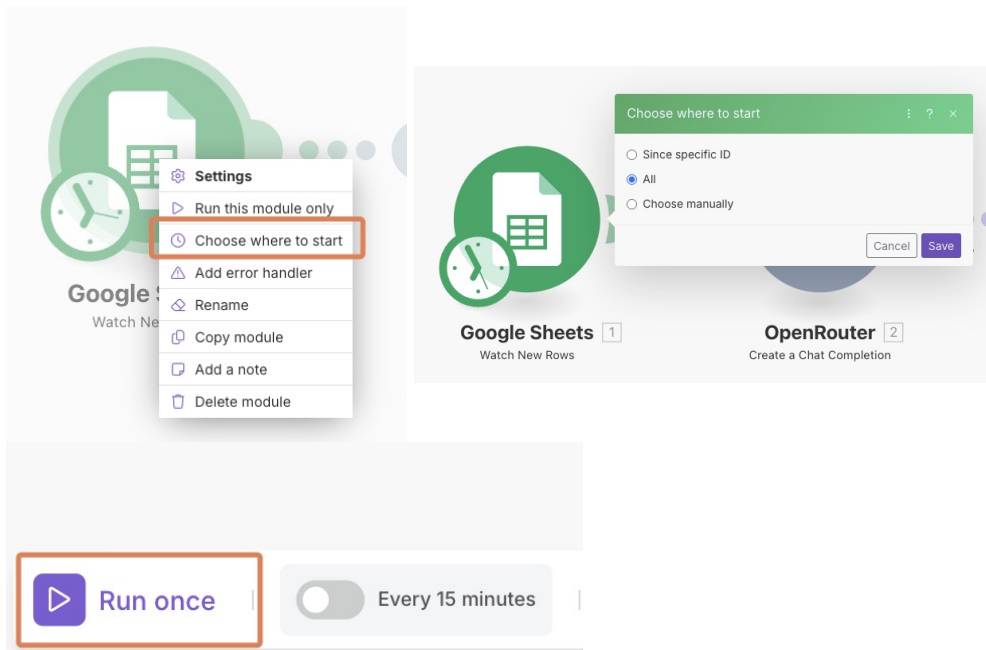
☐ Show advanced settings

## Test the Workflow

Now to test the connection, you need to perform the steps below.



1. Right-click on the Google Sheets module
2. Click **“Choose where to start”**
3. Click **“All”** → **“Save”**
4. Click **“Run once”** at the bottom left



Note that you will need to do this every time you want to test your workflow.

If you get no errors, the workflow is working properly.

## Save your work

---

Make does not save your work automatically. You need to click the **Save** icon at the bottom after major changes so that in case you refresh the page or lose connection, you do not lose your work.



## Part 3: Adding AI Research Capabilities

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### Set Up OpenRouter AI Module

---

1. Click the **+** button next to the Google Sheets module
2. Search for **OpenRouter** and select **Create a chat completion**



3. Click **Create a connection** and enter the openrouter API key that was emailed to you some time ago.

## Configure the AI Research Prompt

1. Under **Messages**, click **Add item**

2. In the **Content** field, paste a prompt that looks something like:

A potential client has filled out a form on my website requesting generative AI consulting services. Research the company/organization below and create a clear, elegant markdown report that covers an introduction to the business as well as potential generative AI opportunities for their work.

Use emojis, tables, and lists for visual appeal while keeping your writing clear and concise.

If you cannot find online information about the company, just indicate so.

Do not include any prelude or postlude.

Company Name and Geography:

3. After the prompt, click the variable selector and choose the company name field from your Google Form (e.g., "What is your company name...")

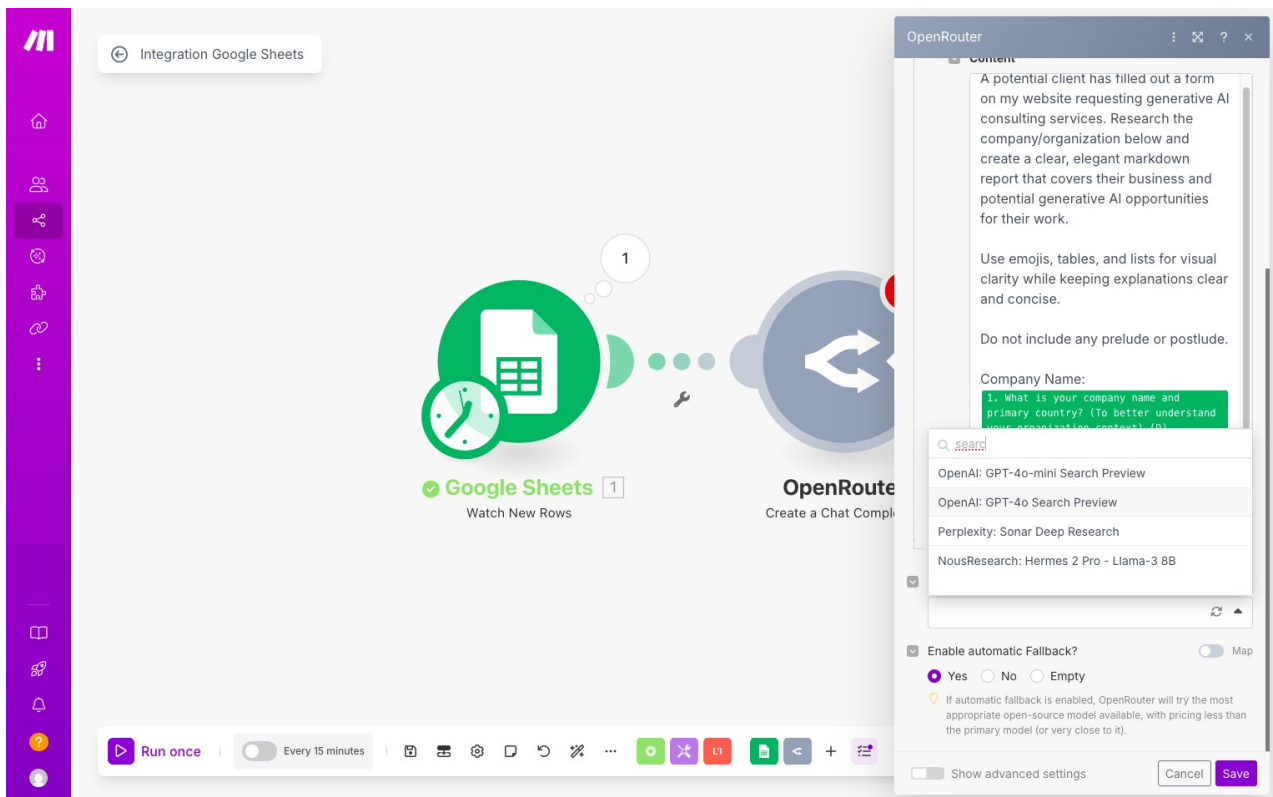
So each time someone fills out the form, the company name will be added to the prompt, and send to the AI model.

4. Set the **Role** field to **User**

The screenshot displays the OpenRouter integration setup interface. The main window shows a 'Google Sheets' integration with a 'Watch New Rows' trigger. A modal window titled 'OpenRouter' is open, showing the 'Content' field with the prompt text. The 'Role' dropdown is set to 'User', and the 'Model' dropdown is set to 'System'. The 'Enable automatic fallback?' checkbox is checked, and the 'Show advanced settings' checkbox is unchecked. The 'Run once' button is visible at the bottom left of the main window.

5. In the **Model** field, type “search” and select **OpenAI: GPT-4o Search Preview**

This model can search the internet to research companies, providing more comprehensive and current information. In the past we used Perplexity Sonar for this, but our testing shows that OpenAI’s model works a bit better for this task.



6. Click **Save**

## Test and Save Your Workflow

Now that you have two modules, you should test the workflow again!

1. As you did before, right-click the Google Sheets module → **Choose where to start** → **All** → **Save**
2. Then click **Run once** at the bottom left to test the entire workflow
3. If you get no errors, then all is working fine.
4. Click the Save icon in the bottom toolbar to save your work.

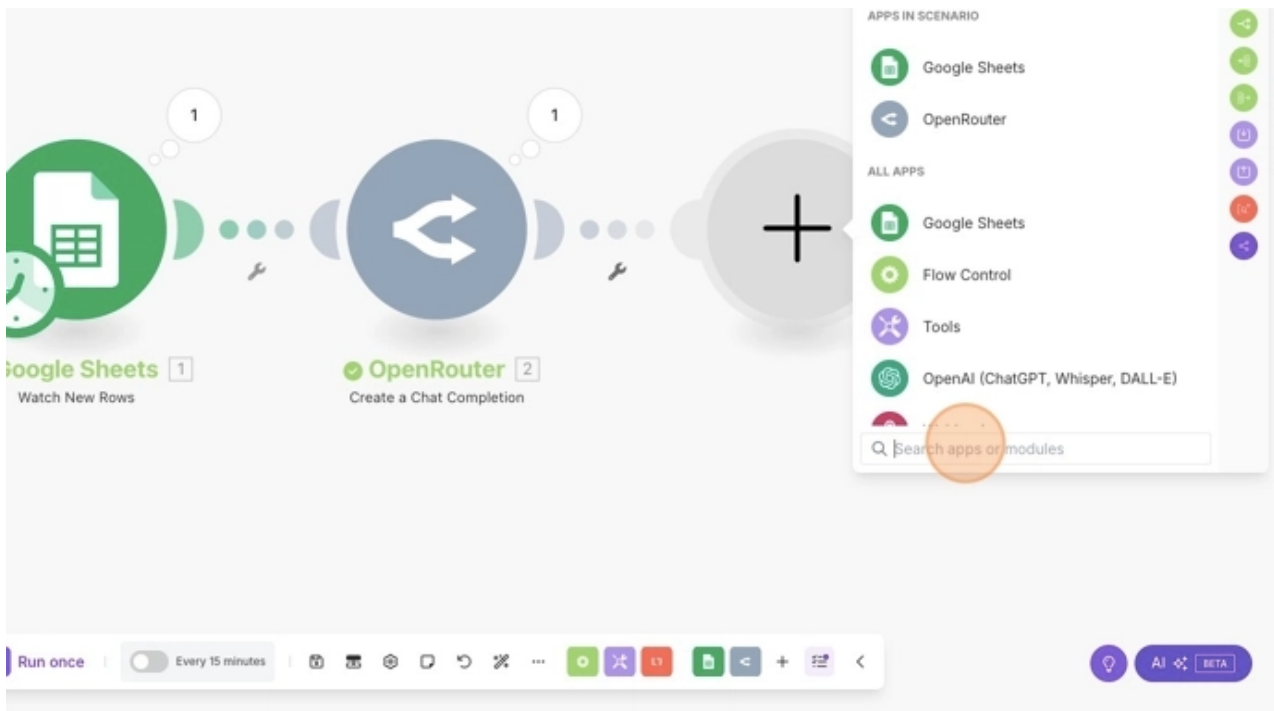
## Part 4: Converting to HTML

By default, OpenAI’s Search model outputs a format of content called “markdown”. But Google Docs prefers the HTML format, so we can use a built-in Make.com module to convert from markdown to HTML.

### Add Markdown to HTML Converter

1. Click the **+** button next to the OpenRouter module
2. Search for **Markdown** and select **Markdown to HTML**

3. In the **Markdown** field, select the AI response from the variable picker:  
Navigate to OpenRouter variables → **Choices** → **Message** → **Content** and click on **Content**.
4. Set **Sanitize** to **Yes**
5. Click **Save**



## Test And Save the Workflow

Again, follow the sequence to test your flow:

Right-click the first module → **Choose where to start** → **All** → **Save** → **Run once** (You may see a warning about running with transformation as the last node - click **Run anyway**)

Save your work.

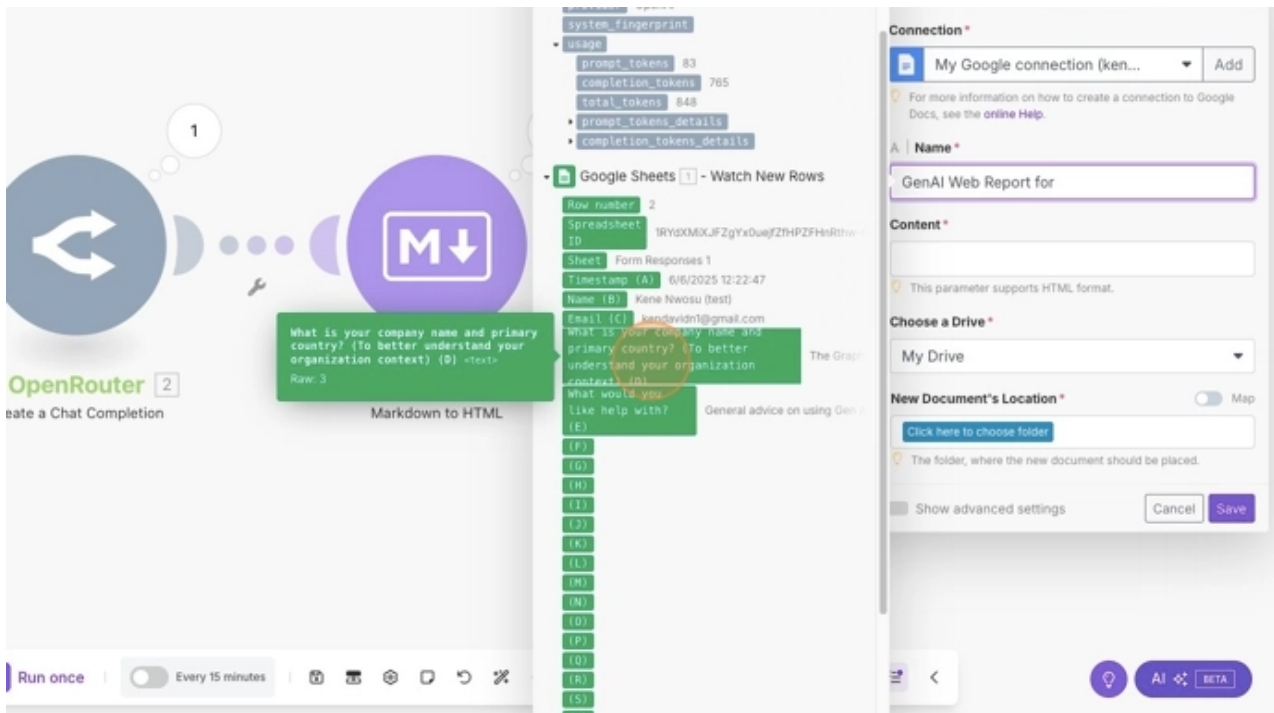
## Part 5: Creating Google Docs Reports

### Set Up Google Docs Module

1. Click the **+** button to add another module
2. Search for **Google Docs** and select **Create a document**



3. For the document **Name**, type: **GenAI Web Report for** then select the company name variable from your form
4. For the **Content** field, we want to add the HTML output from the Markdown converter. However, for Google Docs to recognize the content as HTML, we need to wrap the HTML in a **<html>** tag. So:
  - Type: **<html>**
  - Add the **HTML** variable from the Markdown converter
  - Type: **</html>**
5. Under **New document's location folder**, search for and select your "GenAI Consultations" folder
6. Click **Save**



## Test and Save Workflow

Run your complete workflow test again and save your work. If successful, check your Google Drive for the newly created report document. If you're unhappy with the report, try editing the prompt and rerunning your test until you get an output that works.

## Part 6: Adding Email Notifications

### Set Up Resend Email Service

Sending emails from your automations is often useful. In an ideal case, you can connect your own email directly to the make.com service and draft and send emails from your account using your automations.

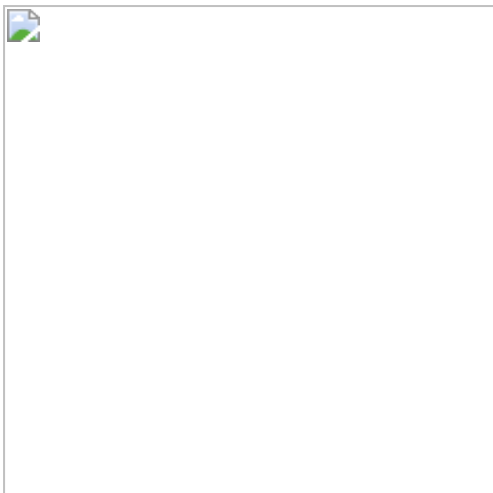
But connecting personal Gmail accounts to Make.com requires a somewhat long setup process (see: [https://www.youtube.com/watch?v=VIDT\\_D9zjnU](https://www.youtube.com/watch?v=VIDT_D9zjnU)) and connecting company email accounts typically requires special permissions.

To simplify our demonstration, we'll instead use a service called Resend.com (email service for app development) for easy email delivery:

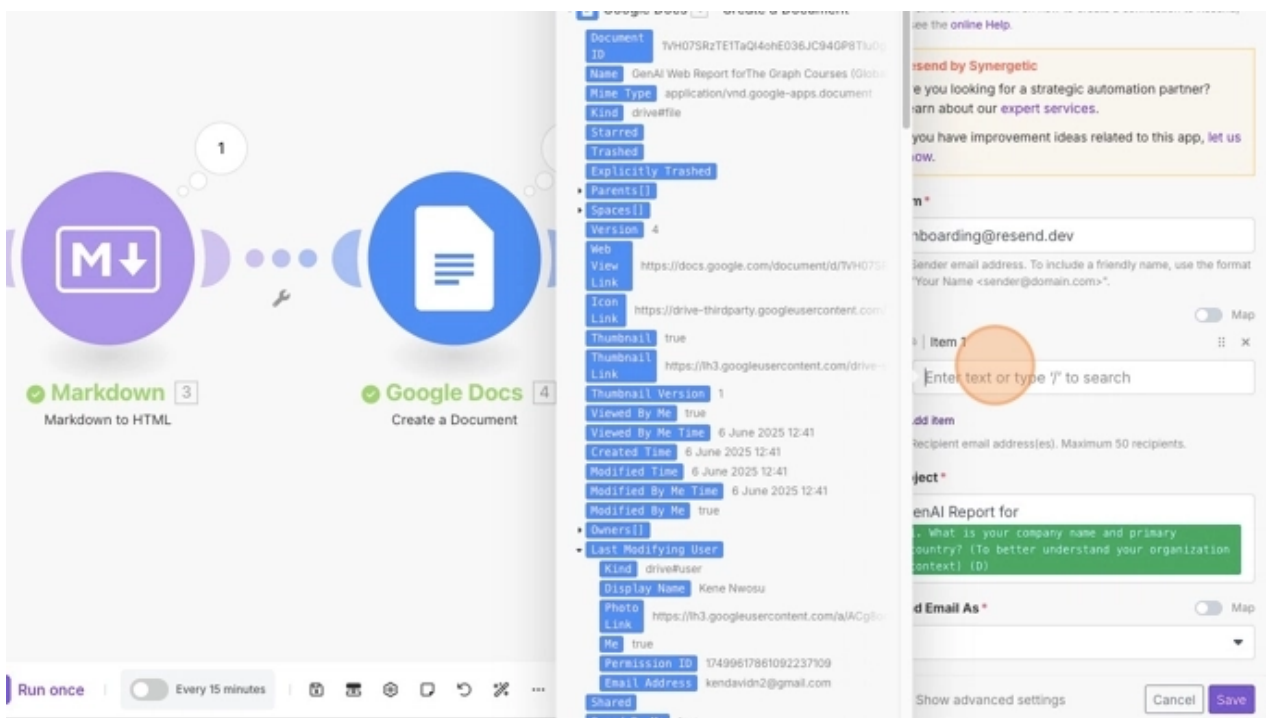
1. Go to [resend.com](https://resend.com) and create a free account.
2. On the homepage or API Keys page, create an API key and copy it.

### Configure Email Module in Make.com

1. Add a new module and search for **Resend**
2. Select **Send an Email**



3. Click **Creating a connection...** and paste your Resend API key
4. Click **Save**
5. Configure the email settings:
  - **From:** `onboarding@resend.dev`
  - **Subject:** Type `GenAI Report for` then add the company name variable
  - **To:** Enter your email address
  - **Send email as:** Select **HTML**
  - **HTML content:** Add the HTML variable from the Markdown converter
6. Click **Save**



## Test Email Delivery

Run your complete workflow test. You should receive an email with the generated report. Save your work.

## Part 7: Activating Your Automation



Congratulations! You've now built a complete automation that will process new form submissions automatically! Let's make it live.

## Set Schedule and Activate

---

1. Locate the scheduling section next to the **Run once** button
2. We recommend changing the frequency from **Minutes** to **120** (runs every 2 hours)  
This prevents using too many operations on the free plan (1000 credits limit)
3. Click **Save**
4. Toggle the scenario **On** or click **Activate scenario** if prompted



Your automation is now live and will process new form submissions automatically!

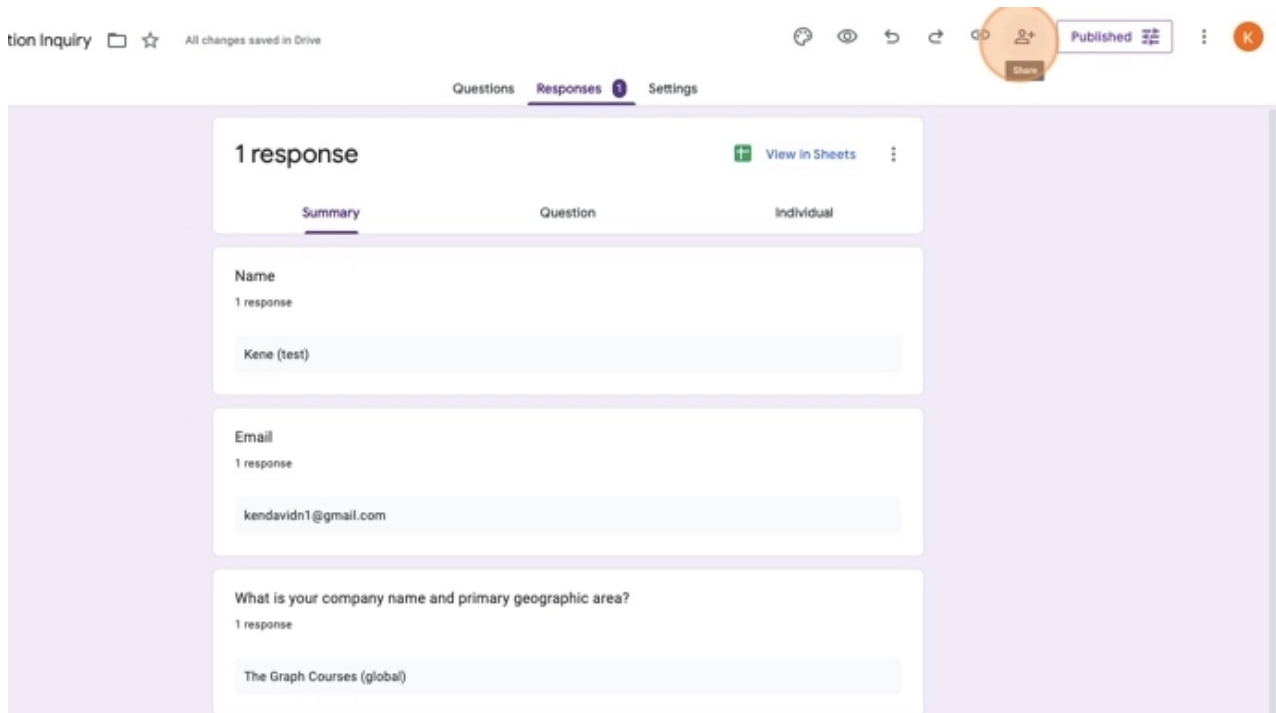
## Part 8: Embedding the Form in Your Website

---

### Get the Embed Code

---

1. Return to your Google Form
2. Click the three-dot menu (⋮) in the top right
3. Select **Embed**
4. Copy the provided HTML embed code



## Add to Your Portfolio Website

1. Go to ChatGPT or your favorite LLM and open your existing portfolio conversation, or copy paste in the index.html code from your page.
2. Provide a prompt that looks something like this:

Please update my portfolio website to include this consultation form.  
Add a new section called "Contact for Consultation" and embed this Google Form:

[Paste your embed code here]

3. Update your GitHub Pages website with ChatGPT's new HTML code
4. Commit/Push the changes to make your consultation form live

## Submission

To submit your work:

1. Export and upload your scenario blueprint. In the bottom toolbar, under the more options, click **Export Blueprint**. This contains a file that represents your completed workflow. (This means if you wanted you could share, [or sell!] your workflows!)
2. Upload that file to the submission form.

# Workshop 6: GenAI-Assisted Design

---

 [the-graph-courses.github.io/aiw\\_materials/week\\_06\\_workshop/week\\_06\\_workshop\\_instructions.html](https://the-graph-courses.github.io/aiw_materials/week_06_workshop/week_06_workshop_instructions.html)

The Graph Courses

In this session you will try two small but practical GenAI design workflows:

1. **Create a promotional flyer** for our course showcase using **Ideogram** for image generation and **Canva** for final editing. (You're encouraged to embed this on your website, and share this flyer with colleagues and friends to get them to come to the showcase.)
2. **(Optional challenge) Turn a research paper into a visual abstract** using **Claude** and **HTML** or **SVG**.

## Task 1 – AI Flyer for the Final Presentation

---

**Account Required:** You'll need a free **Ideogram.ai** account. Sign up at **ideogram.ai** if you haven't already.

### 1-A Generate the first draft in Ideogram

---

1. Go to **ideogram.ai** and log in to your account. (We're using ideogram instead of ChatGPT's image generation feature because the free image generation on ChatGPT is quite slow and unreliable.)
2. In the prompt box, describe the image you want to create. Copy and paste the following recommended prompt, making sure to replace **[YOUR WEBSITE NAME HERE]** with your actual website name:

Create an elegant modern flyer for the following event:

- ▶ Main headline: "Graph Courses – AI Portfolio Showcase Day".
- ▶ Sub-headline: "Generative AI for Work & Research".
- ▶ Date and Time
  - Friday, 27 Jun 2025 3 – 5 PM (GMT)
  - Sunday, 29 Jun 2025 5 – 7 PM (GMT)
- ▶ Sign up instructions: "To sign up, visit my website [YOUR WEBSITE NAME HERE] and open the chat bubble. Tell the AI agent you would like to attend the showcase. It will collect your details and we'll send you a confirmation email."

3. Click **Generate**. Ideogram will create 4 different variations of your flyer. Choose the one you like best and click **Download**.

### 1-B Fix typography in Canva

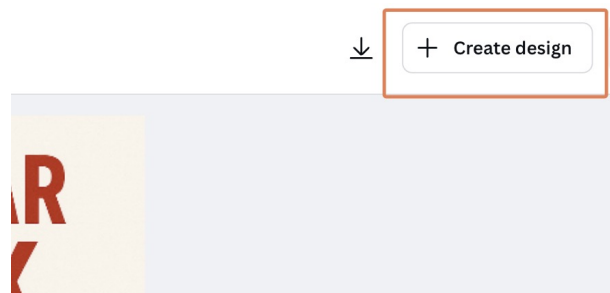
---

**Account Required:** You'll need a **Canva.com** account with pro features. Sign up and start a free trial at **canva.com**. **Important:** Cancel the trial within your account settings immediately after completing the signup process to avoid charges. (If you have already

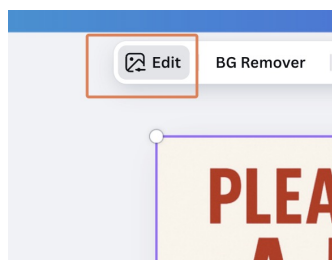
used up your free trial, please create a new account with a different email address. If this is not possible, please contact an instructor for help.)

AI-generated text in the images from Ideogram are likely to be error-ridden. To fix this, we can use Canva's Grab Text feature to make the text editable and fix the errors.

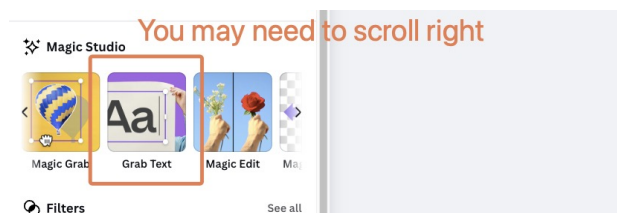
1. Go to **canva.com** and log in to your account.
2. Drag-and-drop your downloaded Ideogram image into the Canva interface
3. Once the image is uploaded, click on it to select it, then click "Create design" at the top right of the screen.



4. With the image still selected, click the top left "edit" button.



Then click on Grab Text under Magic Studio. You may need to scroll right to see it.



5. Use the text tool to correct any typos, change fonts, and adjust colors as needed.

## 1-C Share & Upload

1. Click on "Share" at the top right of the Canva page.
2. Click on "Download", change the file type to "PDF Standard" and click "Download".
3. Upload the PDF to your assignment submission area on the course page.

## Optional Challenge – AI Visual Abstracts

**Account Required:** You'll need a free **Claude.ai** account. Sign up at **claude.ai** if you haven't already.

In this section, we'll use AI and design tools to turn a research paper into a visual abstract. A visual abstract is a pictorial representation of the main findings of a research paper, and is a great way to communicate the key results of a paper in an appealing visual format.

## 2-A Choose Your Paper

---

Pick any recent research paper you are interested in. Make sure you have access to both the paper title and abstract text, as you'll need to copy and paste these into Claude in the next step.

## 2-B Generate Visual Abstract with Claude

---

Go to **claude.ai** and log in to your account. Use Claude to create either an SVG or HTML-based "visual abstract" of your chosen paper.

(We recommend Claude over ChatGPT for this task because - a) it tends to be better at design tasks, and - b) it gives you an easy way to download SVG code.

Choose one of the following two approaches (SVG or HTML) and give Claude the corresponding prompt. Feel free to try both approaches and compare the results.

### Option 1: SVG Visual Abstract

Create an SVG code-based visual abstract of the following paper:

[Paper Title]  
[Paper Abstract]

### Option 2: HTML Visual Abstract

Create an HTML visual abstract of the following paper:

[Paper Title]  
[Paper Abstract]

After Claude generates the initial design, you can ask it to make specific changes or improvements to refine the visual abstract.

## 2-C Edit and Finalize

---

### If you chose SVG:

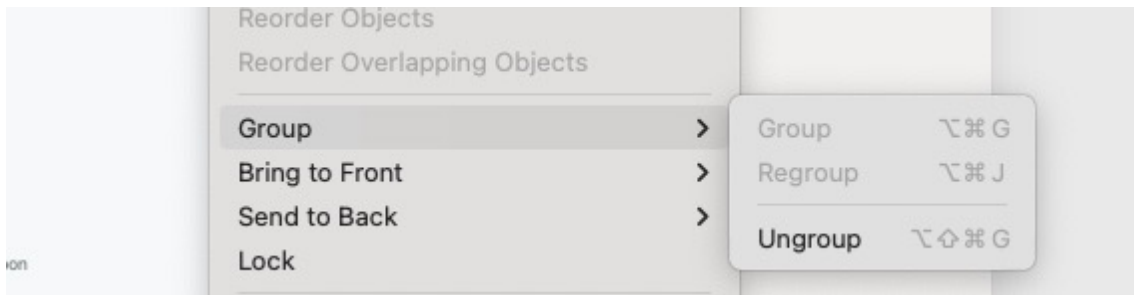
1. In Claude, click on the dropdown arrow at the top right of the SVG preview
2. Click on "Download SVG"



Download SVG in Claude

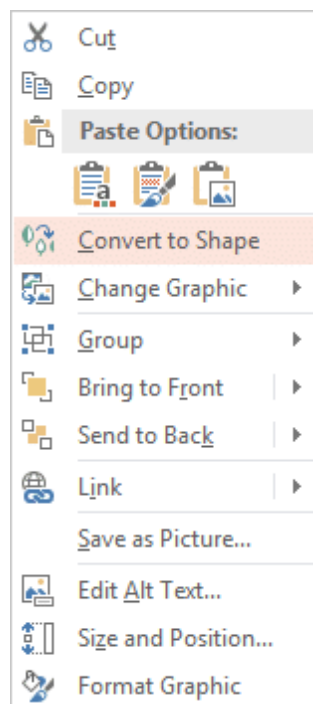
3. Open PowerPoint and drag the downloaded SVG file into a slide
4. To make the elements editable, you need to ungroup/convert the SVG:

**For Mac users:** Right-click on the image, click on “Group”, then “Ungroup”. You may need to repeat this process several times until there’s nothing left to ungroup.



Ungroup in Powerpoint for Mac

**For Windows users:** Right-click on the image and click on “Convert to shape”.



Ungroup in Powerpoint  
for Windows

5. Now you can make any edits you need to individual elements
6. When finished, export as a PDF or take a screenshot of your slide

**If you chose HTML:**

1. Copy the HTML code from Claude's response
2. Go to **grapesjs.com** (a free visual HTML editor) and open the editor
3. Paste the HTML code into the editor
4. Use the visual interface to edit the design as needed
5. For detailed guidance on editing with GrapesJS, watch this video:
6. When you're satisfied with your design, take a screenshot of the final result.

## 2-D Submit Your Work

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Upload your final visual abstract (PDF, screenshot, or exported file) to your assignment submission area on the course page. The page should allow multiple submissions. (After you upload one file, simply repeat the process to upload another file.)

# Final Project: Generative AI for work & research

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 [the-graph-courses.github.io/a iw\\_materials/final\\_proj/final\\_project\\_instructions.html](https://the-graph-courses.github.io/a iw_materials/final_proj/final_project_instructions.html)

The Graph Courses

For your final project, you will design and present an enhanced AI solution relevant to your work or research area, building on one or more modules from this course. You will deliver a **4- minute live presentation** and submit your slide deck.

We recommend that your project should be based on one or more of the core themes covered in each module of the course:

- **A website** (Week 1)
- **A data analytics product** (Week 2)
- **A research/text analysis product** using LLMs (Week 3)
- **A chatbot/agent** (Week 4)
- **An automation workflow** (Week 5)
- **A design/communication campaign** (Week 6)
- **Or a combination of some of these**

The presentation will be on either Friday, 27 Jun 2025 3 – 5 PM (GMT) or Sunday, 29 Jun 2025 5 – 7 PM (GMT). You can pick whatever session works best for you.

## Project Requirements

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- **Relevance:** The project should address a real-world problem or opportunity in your work, research, or study area.
- **AI Integration:** Use at least one GenAI tool or workflow covered in the course.
- **Originality:** Enhance or extend the example from the chosen week(s) with your own data, content, or workflow.
- **Presentation:** Prepare a clear, engaging 4-minute presentation and submit your slide deck (PDF, PPTX or HTML).
- **Reflection:** Reflect on what worked and what was not ideal, especially with regard to AI limitations and or ethical considerations.

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## Instructor Feedback Sessions (June 20, 22)

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Our second-to-last session will be an optional session where you can get feedback and mentorship on your final project. Try to make sure you have at least a draft of the presentation by then.

## Submission Instructions

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1. **Prepare your slide deck** (PDF, PPTX or HTML) We recommend that your presentation should include the following sections, with these recommended (approximate) durations:
  - The problem or opportunity addressed (30 seconds)
  - Your solution and workflow (Live demonstration, screenshots or short video) (60 seconds)
  - Key features and AI tools used (30 seconds)
  - Discussion of current AI model limitations in your solution (30 seconds)
  - Opportunities for improvement as AI models advance (forecasting future capabilities) (30 seconds)
2. **Present live** in your assigned session (4-5 minutes)
3. **Upload your slide deck** to the submission page which will be posted.

## Grading Rubric

| Criteria                           | Excellent (4)                                                        | Good (3)                                  | Satisfactory (2)                         | Needs Improvement (1)           |
|------------------------------------|----------------------------------------------------------------------|-------------------------------------------|------------------------------------------|---------------------------------|
| <b>Relevance</b>                   | Highly relevant to work/research; clear real-world value             | Relevant; some real-world value           | Somewhat relevant; value not fully clear | Not relevant or unclear value   |
| <b>AI Integration</b>              | Creative, effective use of GenAI tools; integrates multiple features | Good use of GenAI tools; some integration | Basic use of GenAI tools                 | Minimal or unclear use of GenAI |
| <b>Originality/Enhancement</b>     | Strongly extends or customizes course example                        | Some extension/customization              | Minor changes to example                 | Direct copy, little change      |
| <b>Presentation Quality</b>        | Clear, engaging, well-structured; visuals support message            | Clear and organized; some visuals         | Adequate; some unclear points            | Disorganized or unclear         |
| <b>Reflection &amp; Next Steps</b> | Insightful reflection; clear next steps                              | Some reflection; next steps mentioned     | Basic reflection                         | Little or no reflection         |

## Some project ideas

- Build a website for a project, course, or lab group. Use LLMs for content generation and structuring the site/documents.
- Develop a comprehensive data dashboard for your field, including multiple visualizations and insights

- Create a knowledge synthesis project on a topic of your choice. Use Elicit or other research tools for comprehensive text analysis
- Build a custom chatbot to support a specific task, such as answering questions about a research paper, guiding a user through a protocol, or managing a project.
- Design and implement an automation addressing a real pain point in your work or organization
- Create a suite of professional communication materials for a campaign for an event, project, or publication (e.g., flyer, social media posts, visual abstract)

## Frequently Asked Questions

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### What if my project is not related to work or research?

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This is fine! It can be related to your personal life, a hobby, or a project you are working on for fun.

### How do I prove I used an AI tool during the project?

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“Work on problems that used to be impossible, but are now really hard.” — Kevin Scott, Chief Technology Officer, Microsoft (2024)

While lots of people are trying to prove they *didn't* use AI in their work, we actually want to see that you *did* use it! This can be tricky to show, especially for things like data analysis where you could have done it all yourself. Here's how to handle this:

1. **Keep Track:** Take some screenshots or short videos of you working with AI tools, like keeping receipts to show what you did!
2. **Show the Difference:** Show how it helped you do more than you could have done on your own. The CEO of Microsoft, Kevin Scott, said this: “Work on problems that used to be impossible, but are now really hard.” Apply this to your project. Build a project that you would not have been able to achieve (at least not within two weeks) without the help of AI tools.

### Can I use AI tools not covered in the course?

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Yes! While you should use at least one tool covered in the course, you're encouraged to explore additional AI tools that enhance your project.

### What if my project doesn't work perfectly?

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That's completely fine and expected! Part of the reflection component is discussing what didn't work well. Failed experiments and limitations are valuable learning experiences.

### Can I present preliminary results or a work-in-progress?

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It's okay if your work is not finished by the end of the course. Be clear about what's preliminary and what your next steps would be.

### How strict is the 4-minute time limit?

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We'll aim for 4-5 minutes total to allow for brief Q&A. Practice your timing to ensure you cover all key points effectively.

### Can I use the same data/project from another course?

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Yes, but you must add significant AI components that weren't part of your original work. The focus should be on the AI enhancement/integration.

### **What if I miss the feedback session?**

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The feedback sessions are optional but highly recommended. If you can't attend, you can reach out via email or during regular office hours for guidance.